

Sumit De

Data Scientist

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SUMMARY

- Data Scientist with Around 5 years of experience leveraging Artificial Intelligence (AI), Deep Learning, Machine Learning, Data Mining, Data Visualization, and Natural Language Processing (NLP) to solve complex business problems and drive impactful results.
- Leveraging deep learning expertise in implementing Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) with LSTMs, and Transformers to tackle complex problems.
- Skilled in using Python and R libraries like NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow, Ggplot2, PyTorch, and SparkNLP.
- Participated in the entire data science project life cycle, actively engaging in phases including data extraction, data cleaning, statistical modeling, and data visualization with large datasets of structured and unstructured data.

SKILLS

- **Languages and IDE's:** Python, R, SQL, Jupyter Notebook, Google Colab
- **Machine Learning:** Linear, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, SVM
- **Deep Learning:** CNN, ANN, RNN, LSTM, LangChain, Hugging Face Transformers (BERT, GPT-3)
- **AI Technology:** Natural Language Processing (NLP), Large Language Model (LLM), GenAI
- **Cloud/Visualizations:** AWS (S3, RedShift, Lambda, Glue, SageMaker), ADLS Gen2, Tableau, Power BI, Excel, Looker
- **Statistical Techniques:** Hypothesis Testing, Decision Trees, Data Visualization, Data Modelling, A/B testing, Model Evaluation
- **Packages and Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Docker, Git, Kubernetes, Keras, NLTK, XGBoost, PyTorch, Kafka
- **Database:** MySQL, PostgreSQL, MongoDB, SQL Server

PROFESSIONAL EXPERIENCE

Charles Schwab, TX | Data Scientist Jan 2024 – Present

- Built machine learning model development with Linear by automating feature engineering and model training, a 30% reduction in development time allowed for quicker iteration and faster delivery of impactful models.
- Improved machine learning model robustness using an innovative approach LangChain to build a data augmentation pipeline, generating synthetic data that enhanced model performance by 20% and addressed data scarcity challenges.
- Engineered and deployed a GAN-based image generation model to create high-quality product images, reducing manual design time by 30% and improving product conversion rates by 15%.
- Led data analysis and predictive modeling initiatives using Python libraries (Scikit-learn, NumPy, pandas) and Spark (MLlib, PySpark), created various advanced customer segmentation and behavior analysis algorithms, resulting in a 25% increase in customer lifetime value prediction accuracy.
- Trained multi-task LLM models for various NLP tasks, including text summarization and named entity recognition (NER), achieving an overall improvement of 15% in model performance across tasks.
- Boosted sales forecasting accuracy by 15% using ARIMA and XGBoost models, providing insights into revenue growth trends visualized in AWS Quicksight.
- Optimized investment portfolios by rigorous Monte Carlo simulations, resulting in a 2.5% increase in annualized returns while managing risk exposure.

Eaton Corporation, PA | Data Scientist Intern May 2023 – Aug 2023

- Led the development of an AI/ML model using a Support Vector Machine (SVM) based on the nature of the problem to detect cavitation in centrifugal pumps, achieving 85.98% precision.
- Ingested and processed 134.27 million data points from 128 files using Python, downsampled the data by 97% for efficient analysis, and applied feature engineering for the selection of relevant features.
- Enhanced the interpretability of model outputs, carried out thorough data visualization and model review. Supported biweekly management decision-making through presentations.

Tata Consultancy Services (TCS), Bengaluru, India | Data Scientist Apr 2021 - Jul 2022

- Established a CNN-based image classification model that achieved 25% accuracy in identifying defects in manufactured products, leading to a reduction in production costs and Streamlined product quality.
- Executed an XGBoost model for time series forecasting, achieving high accuracy in predicting future sales trends, enabling data-driven inventory management decisions, and reducing inventory holding costs.
- Accelerated data preparation by leading to a boost in model accuracy, by designing customized text cleaning and preprocessing tools that effectively handled domain-specific language patterns.
- Optimized the efficiency of legal document review by 15% through an innovative NLP-based solution system that performs named entity recognition and extracts key information, streamlining the legal review process.
- Partnered with cross-functional teams to implement automated dashboards and reporting using Power BI, facilitated stakeholders with data-driven insights, and Increased workforce performance by 20%.
- Implemented scalable and cost-effective data lake using ADLS Gen2, enabling the storage and processing of massive datasets for machine learning model training and real-time analytics, resulting in a 20% reduction in data storage costs.

Zensar Technologies, India | Junior Data Scientist Jan 2019 – Mar 2021

- Developed predictive models using Decision Trees, Random Forests, and XGBoost within the Scikit-learn framework to optimize marketing campaign ROI. Engineered features encompassing customer demographics, purchase history, and campaign metrics. Implemented Random Forest to predict customer response with 72% accuracy.
- Analyzed data and collaborated with the team to create comprehensive reports and presentations that effectively conveyed key metrics and trends using clear data visualizations using Tableau.
- Engineered 100+ distinct features by generating feature intersections, normalizing datasets, and encoding categorical variables with Scikit-learn's preprocessing tools; streamlined model performance in predictive analytics projects.
- Utilized marketing campaign performance through data-driven decision-making. Conducted A/B testing and analyzed results to identify the most effective campaign variations, leading to an increase in click-through rates.
- Initiated anomaly detection models using statistical and machine learning techniques to identify outliers and anomalies in large-scale datasets, leading to a 40% reduction in false positives and an improvement in anomaly detection accuracy.
- Streamlined Data Extraction utilized SQL queries to extract and transform data from legacy MySQL and SQL Server databases, reducing data retrieval time.

EDUCATION

Master of Science in Information Systems May 2024
University of Texas at Arlington, TX GPA: 3.6 / 4.0

PROJECTS

Real-Time Attendance System using Face Recognition, Coursework Project

- Developed prototype model of attendance system using face recognition. Made use of deep learning (CNN) and computer vision for face detection and SVM for face classification. Implemented triplet loss in a face recognition system to ensure embeddings of the same person are closer than those of different people, enhancing identification accuracy. Achieved accuracy of 97.31% with a dataset of 2100 training images and 900 test images.

Airport Analyzer, Business Symposium

- Leveraged Selenium for web scraping, NLP for text processing, LangChain for optimized text chunking, and LLM for data-driven insights, enabling airport authorities to make data-driven decisions based on user reviews. Created a user interface using Streamlit. Implemented a RAG approach to optimize costs by 70%. Secured 2nd place among 50+ competing teams across North Texas.